



Data Communication Network Basis



Foreword

- Communication has always been with us ever since the origin of human society. Communication has been playing an increasingly important role especially since human society entered the information era in the 1970s or 1980s.
- The communication mentioned in this course refers to the communication implemented through a data communication network. This course describes the concepts related to communication and a data communication network, information transfer process, network devices and their functions, network types, and typical networking. In addition, this course briefly introduces the concepts related to network engineering and network engineers.



Objectives

- On completion of this course, you will be able to:
 - Understand the concepts related to communication and a data communication network.
 - Be able to describe the information transfer process.
 - Differentiate network devices of different types and understand their basic functions.
 - Understand different network types and topology types.
 - Understand the concepts related to network engineering and network engineers.



Huawei Device Icons



General router



General switch



Core switch



Aggregation switch



Access switch



Stacked switches



Firewall



General NMS



AP



Base station



General server



Cluster



FTP server



Authentication server



PN user



Enterprise network user



Enterprise



Business trip



AC



Wi-Fi signals



Internet



Network cloud 1



Network cloud 2



IP phone



PC



Pad



Mobile phone



Laptop



Contents

- 1. Communication and Networks**
2. Network Types and Topology Types
3. Network Engineering and Network Engineers



Concept of Network Communication

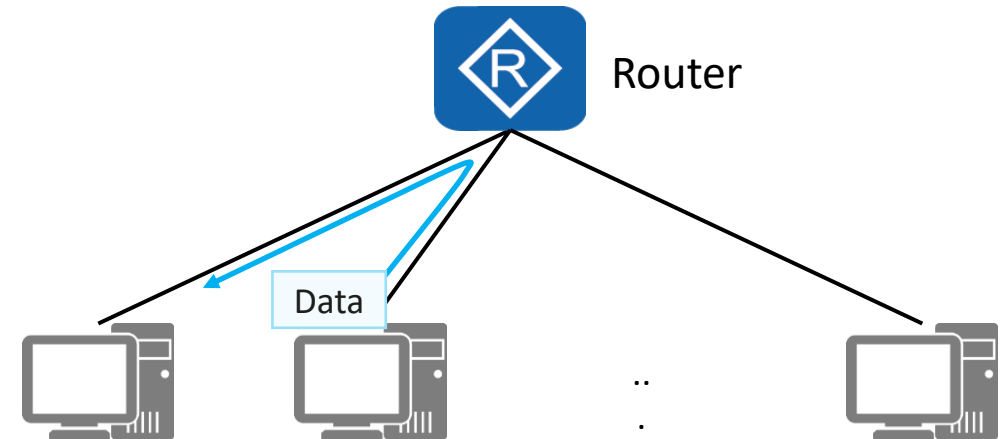
- Communication refers to the information transfer and exchange between people, between people and things, and between things through a certain medium and behavior.
- Network communication refers to communication between terminal devices through a computer network.
- Examples of network communication:



A. Files are transferred between two computers (terminals) through a network cable.



C. A computer (terminal) downloads files through the Internet.

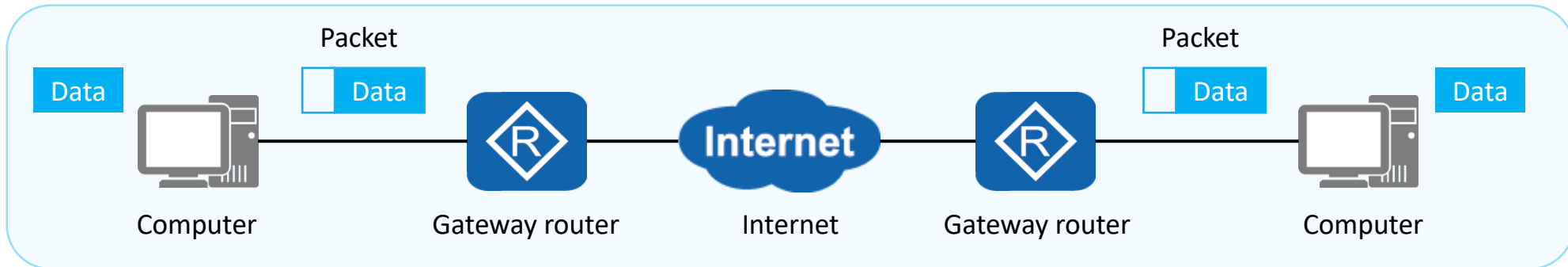
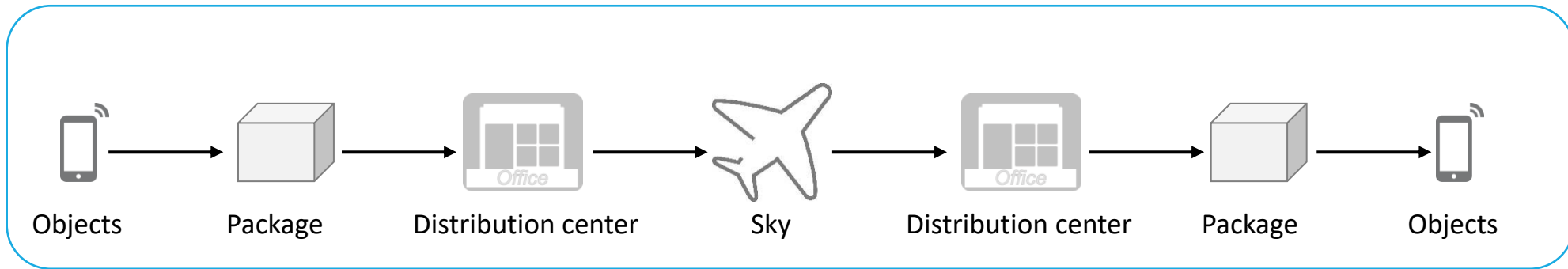


B. Files are transferred among multiple computers (terminals) through a router.



Information Transfer Process

- Virtual information transfer is similar to real object transfer.





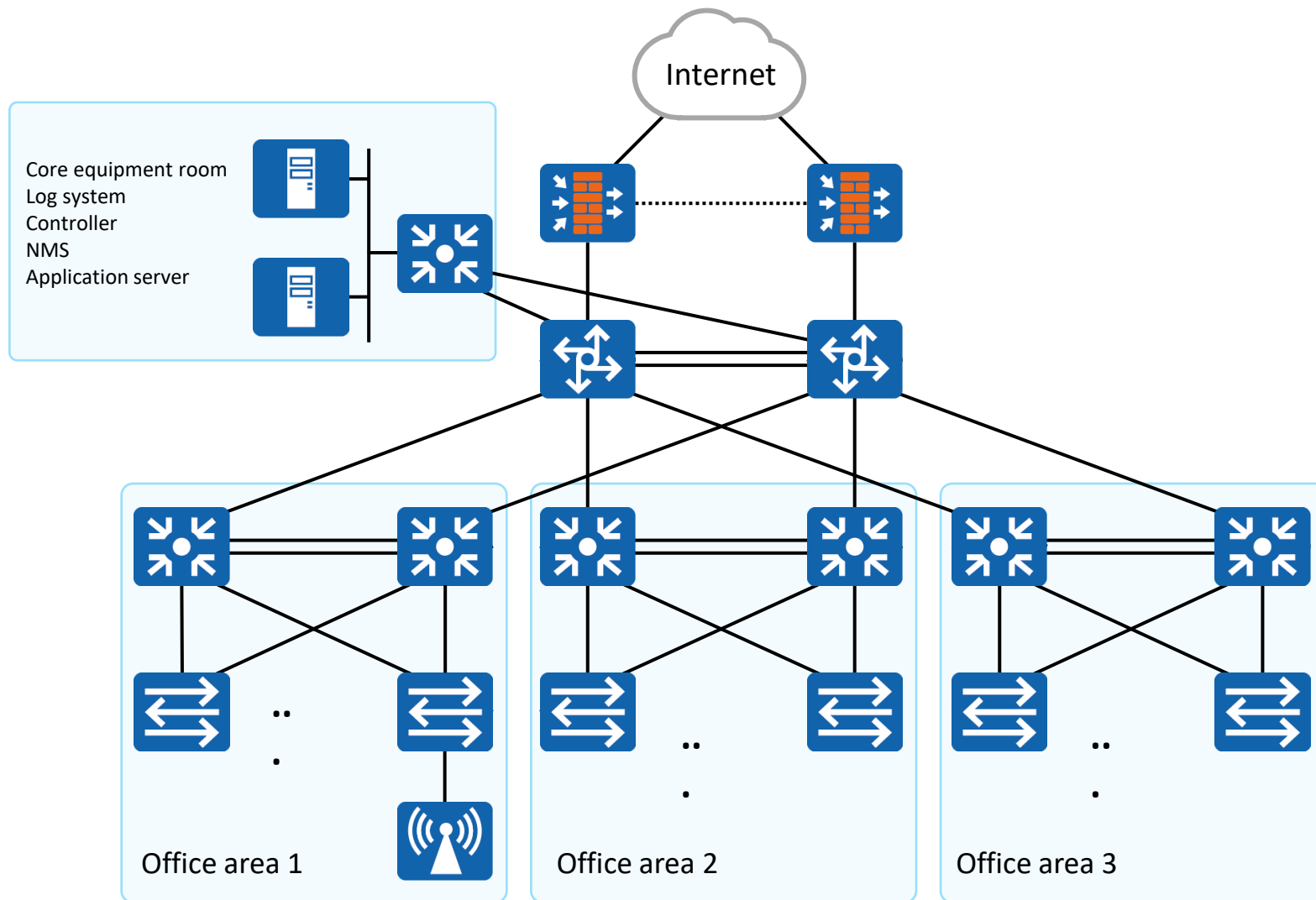
Common Terms

Term	Description
Data payload	Information conveyed
Packet	Data unit switched and transmitted on the network
Header	Information segment added before the data payload
Tail	Information segment added after the data payload
Encapsulation	Process of adding a header and a tail to a data payload to form a new packet
Decapsulation	Process of removing the header and tail from a packet to obtain the data payload
Gateway	Network device that provides functions such as protocol conversion, route selection, and data exchange
Router	Network device that selects a forwarding path for packets
Terminal device	End device of a data communication system, used as a sender or receiver of data



Concept of the Data Communication Network

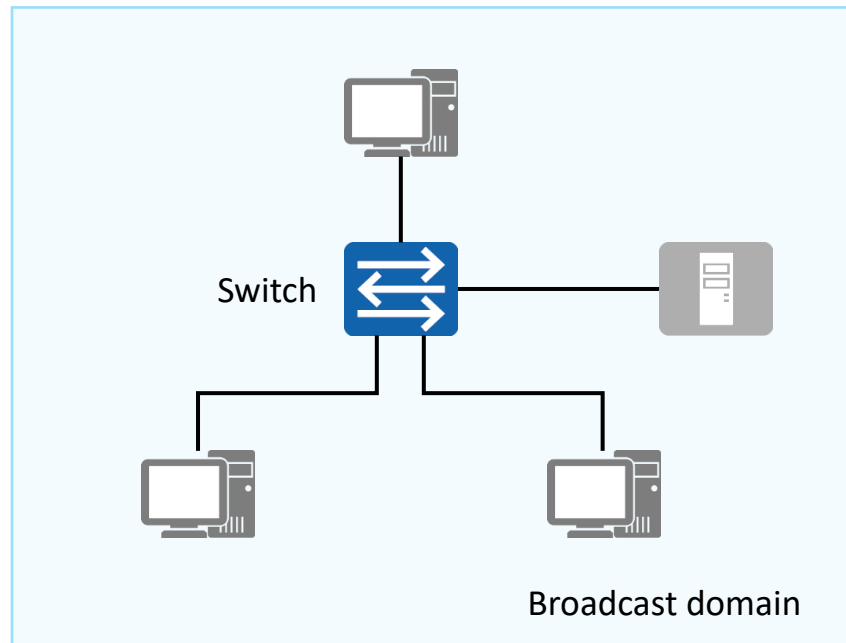
- Data communication network:
Communication network that consists of routers, switches, firewalls, access controllers (ACs), access points (APs), PCs, network printers, and servers
- Function:
To implement data communication





Switches

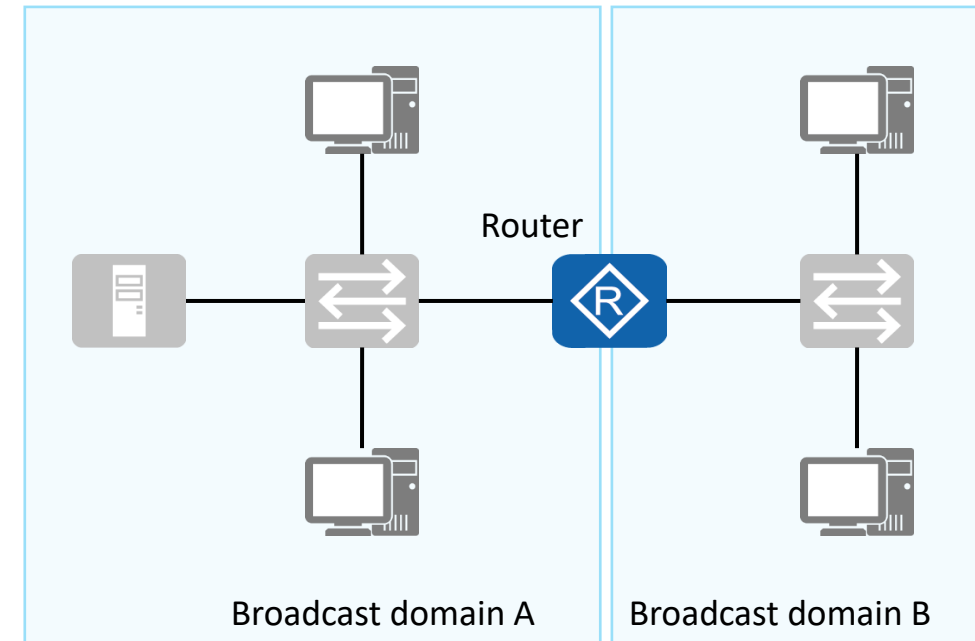
- Switch: a device closest to end users, used to access the network and switch data frames
 - Network access of terminals (such as PCs and servers)
 - Layer 2 switching





Routers

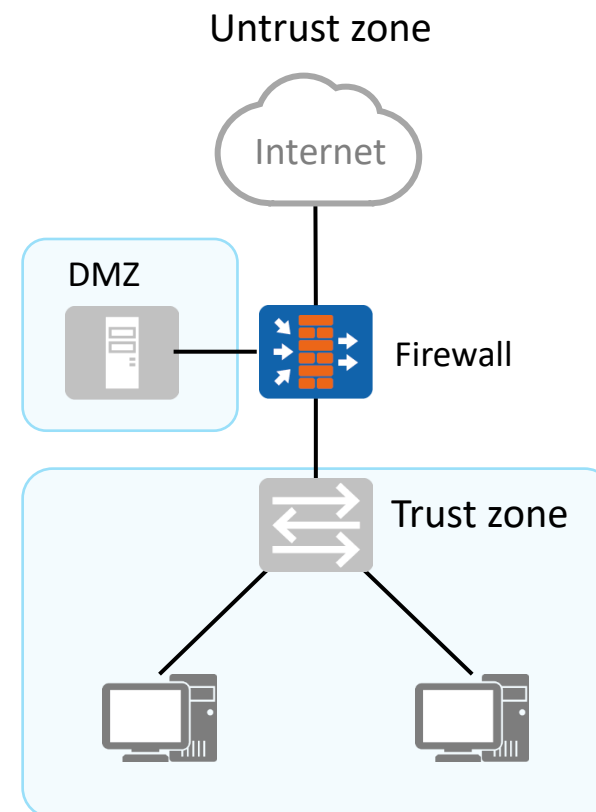
- Router: a network-layer device that forwards data packets on the Internet. Based on the destination address in a received packet, a router selects a path to send the packet to the next router or destination. The last router on the path is responsible for sending the packet to the destination host.
 - Implementing communication between networks of the same type or different types
 - Isolating broadcast domains
 - Maintaining the routing table and running routing protocols
 - Selecting routes and forwarding IP packets
 - Implementing WAN access and network address translation
 - Connecting Layer 2 networks established through switches





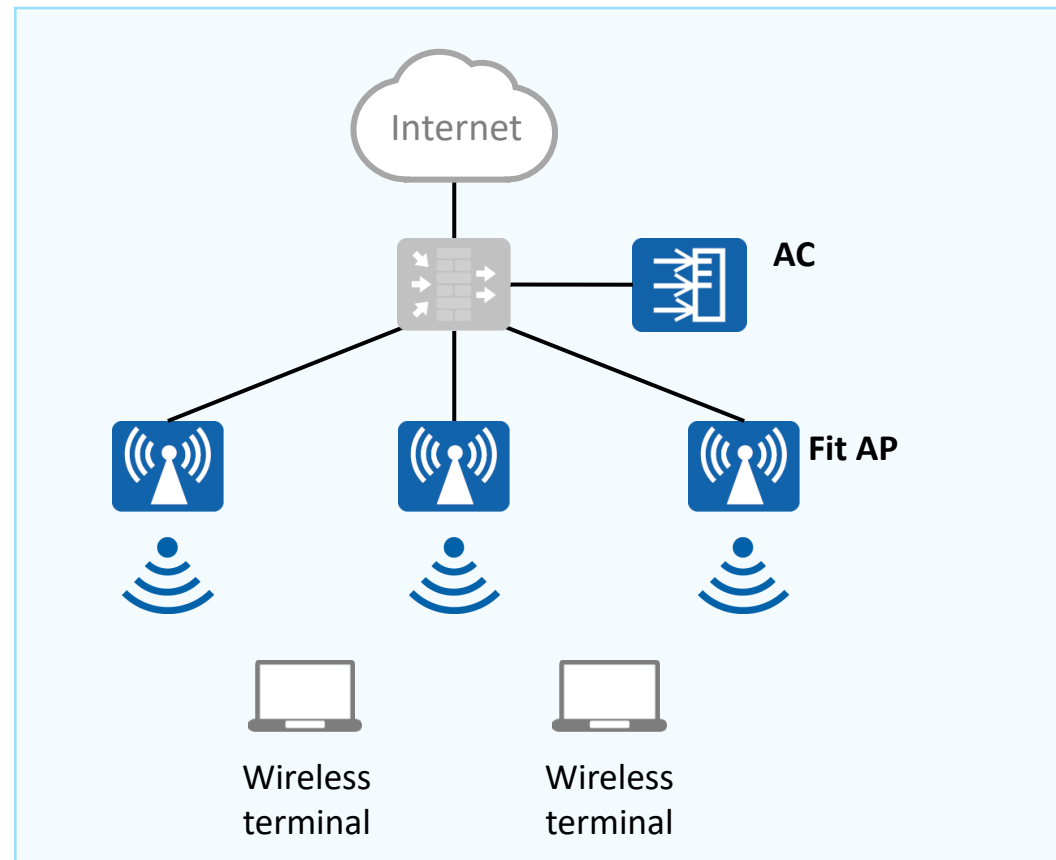
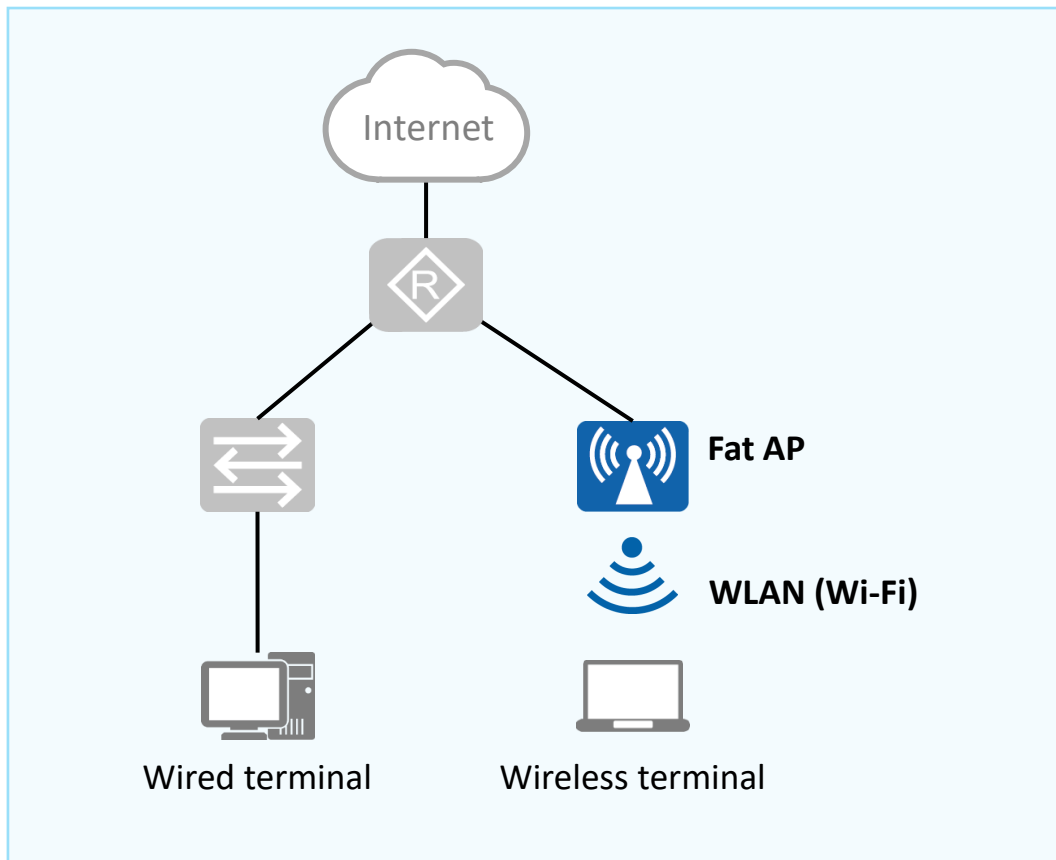
Firewalls

- Firewall: a network security device used to ensure secure communication between two networks. It monitors, restricts, and modifies data flows passing through it to shield information, structure, and running status of internal networks from the public network.
 - Isolating networks of different security levels
 - Implementing access control (using security policies) between networks of different security levels
 - Implementing user identity authentication
 - Implementing remote access
 - Supporting data encryption and VPN services
 - Implementing network address translation
 - Implementing other security functions





Wireless devices





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- 2. Network Types and Topology Types**
3. Network Engineering and Network Engineers

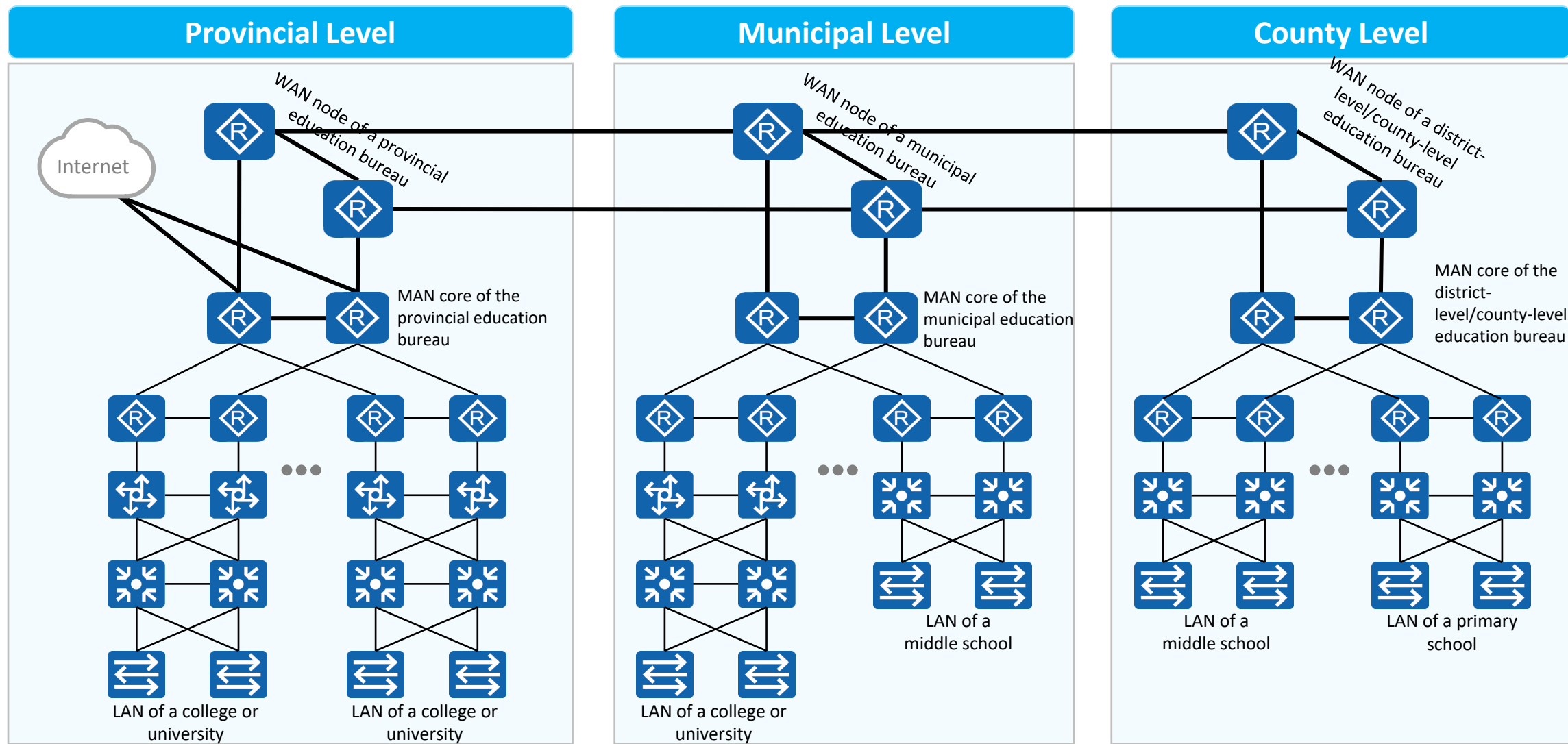


LAN, MAN, and WAN

- Based on the geographical coverage, networks can be classified into local area networks (LANs), metropolitan area networks (MANs), and wide area networks (WANs).
 - LAN
 - A LAN is a network that consists of computers, servers, and network devices in a geographic area. The coverage of a LAN is generally within several thousand square meters.
 - Typical LANs include a company's office network, a cyber bar network, a home network.
 - MAN
 - A MAN is a computer communication network established within a city.
 - Typical MANs include broadband MANs, education MANs, and municipal or provincial e-government private lines.
 - WAN
 - A WAN generally covers a large geographical area ranging from tens of square kilometers to thousands of square kilometers. It can connect networks of multiple cities or even networks of countries (as an international large-scale network) and provide long-distance communication.
 - The Internet is a typical WAN.



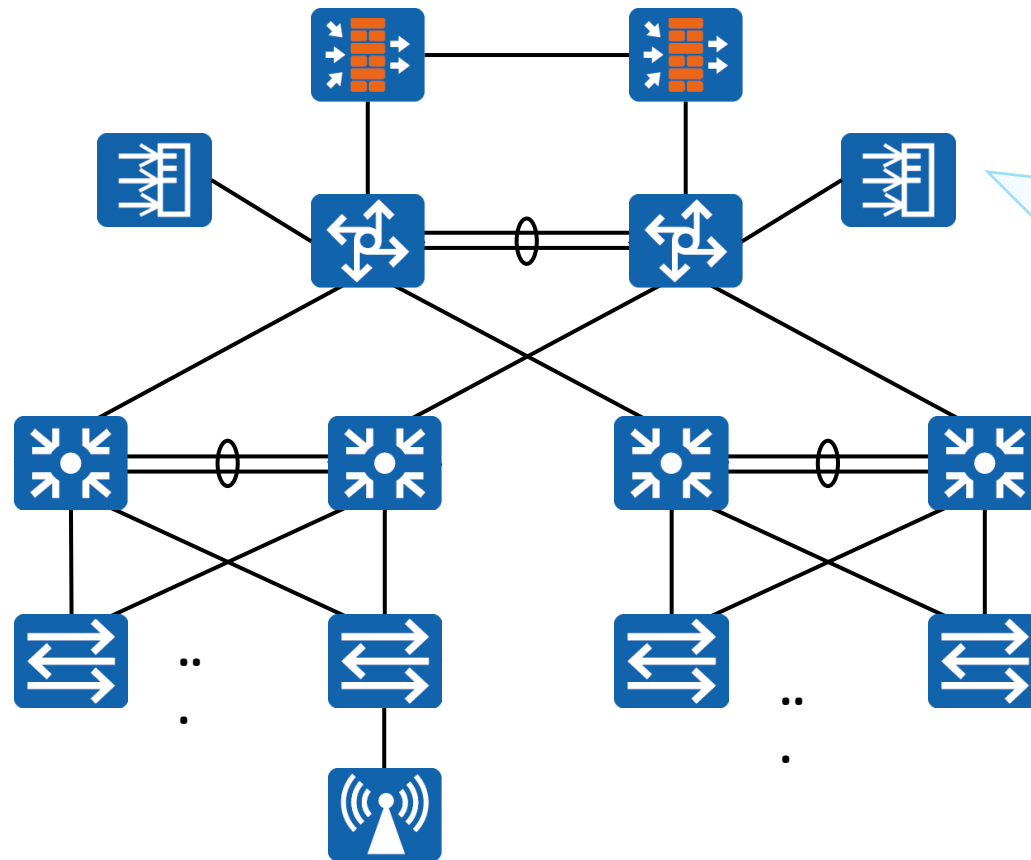
LAN, MAN, and WAN in the Education Industry





Network Topologies

- A network topology is a structured layout presented using transmission media (such as twisted pairs and optical fibers) to interconnect various devices (such as computer terminals, routers, and switches).

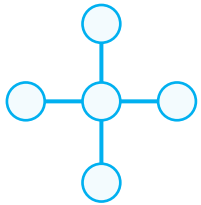


The network topology is used to describe the physical or logical structure of a network in the network engineering field, and is a very important network concept.

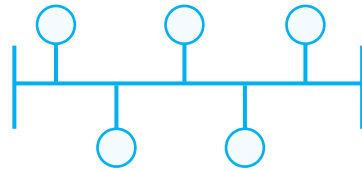


Network Topology Types

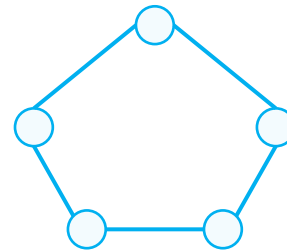
- Network topologies are classified into star, bus, ring, tree, full-mesh, and partial-mesh network topologies.



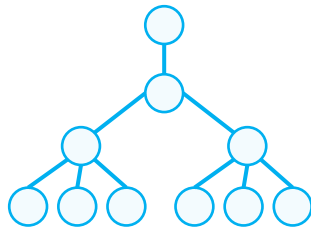
Star network topology



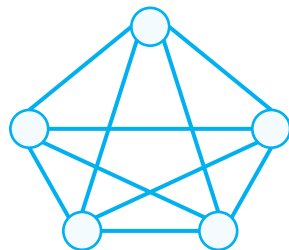
Bus network topology



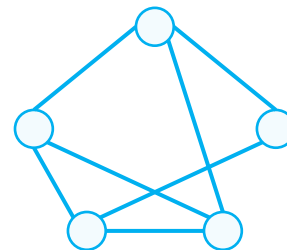
Ring network topology



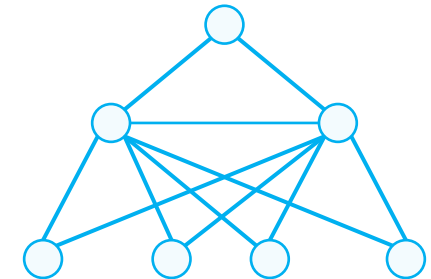
Tree network topology



Full-mesh network topology



Partial-mesh network topology



Combined network topology



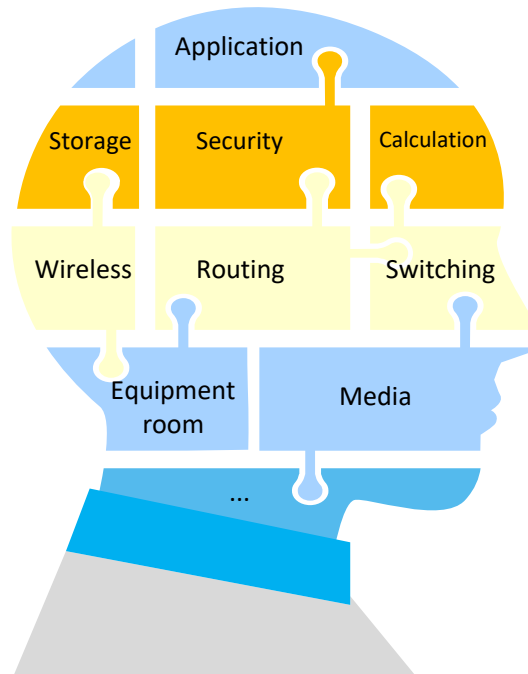
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Network Engineering

- Network engineering
 - Network engineering refers to planning and designing feasible solutions based on network application requirements and computer network system standards, specifications, and technologies under the guidance of information system engineering methods and complete organizations, as well as integrating computer network hardware devices, software, and technologies to form a cost-effective network system that meets user requirements.
- Technical modules covered by network engineering:





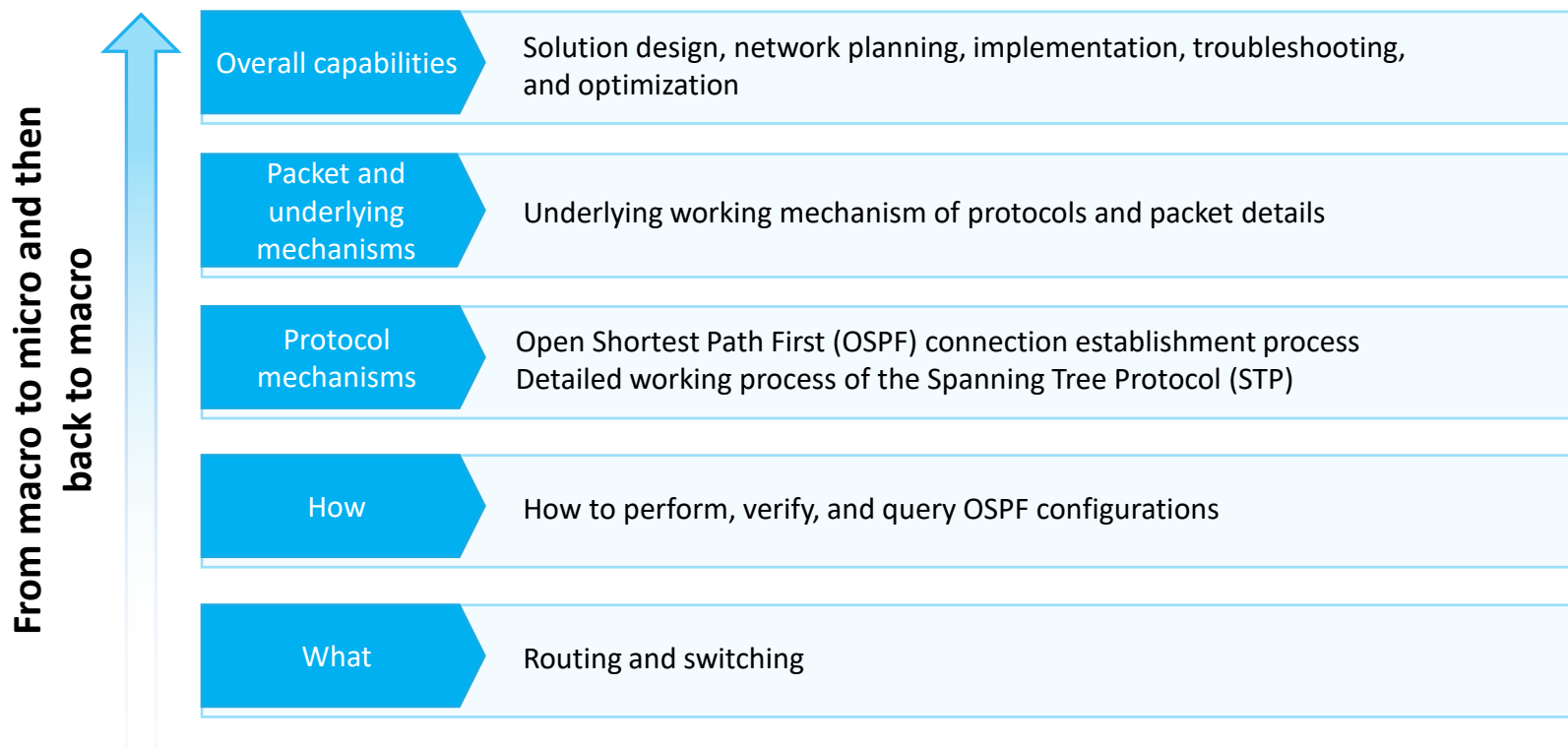
Network Engineer

- Network engineer
 - Network engineers are technology professionals who master professional network technologies, have professional skills, professionalism, and project implementation experience in the network engineering field, and are able to fully communicate with customers or other project stakeholders onsite. In addition, they can develop implementation solutions and project plans (recognized by project stakeholders) based on customer requirements and environment factors, fully mobilize resources of all parties to ensure timely and high-quality project implementation, and provide training for stakeholders and deliver engineering documents after the project is implemented.
- Comprehensive capability models for network engineers:

Process specification	Business etiquette	Team collaboration
Industry knowledge	Values	Business management
Engineering knowledge	Service awareness	Presentation capability
Product knowledge	Information collection	Problem solving
Technical knowledge	Learning competency	Communication capability
Professional knowledge	Basic qualification	Professional skills



Network Engineers' Technology Development Path





Huawei Certification Injects Vitality into Talent Development for Enterprises



Certification exam

- Cultivate experts who understand both business and technologies.
- Cultivate platform construction and service application experts based on HUAWEI CLOUD.
- Focuses on ICT infrastructure and cultivates architecture talent in all ICT fields.



Providing talent with career development paths

- Meets enterprise talent's career evolution from an engineer to a senior engineer, and then to an expert.
- Provides a hierarchical certification system that provides customized talent growth paths in accordance with job-based capability requirements and supports in-depth professional development, integration, and expansion, reducing the talent cultivation cost for enterprises.



Facilitating enterprise innovation and transformation

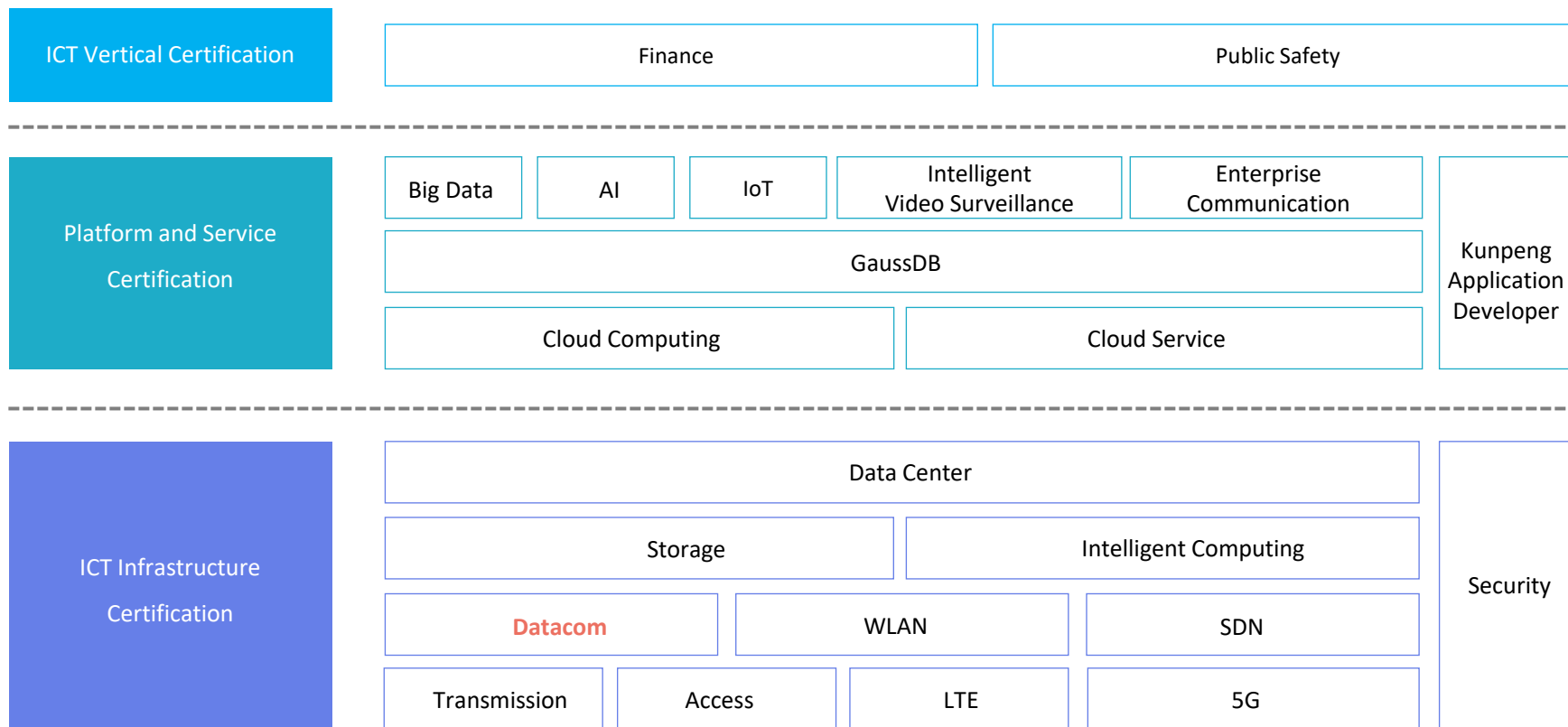
- Provides authoritative certification for ICT talent. The ICT talent with authoritative certification helps ensure project delivery quality and improve customer satisfaction.
- Enhances the overall performance and productivity of enterprises.
- Accelerates business innovation and transformation, and improve the overall operational efficiency.



Huawei Certification Portfolio

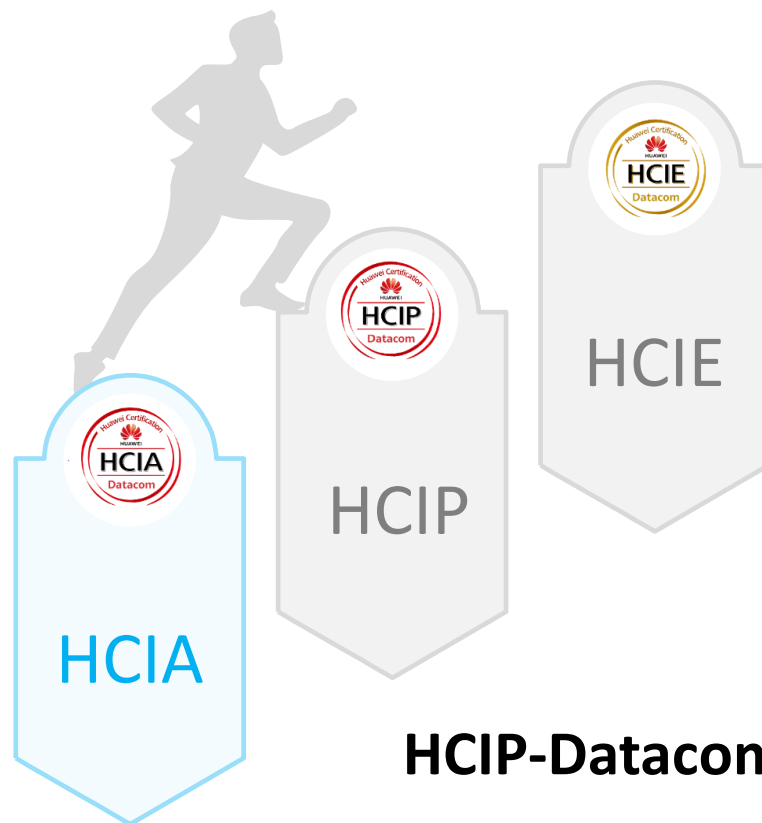
- Huawei certification covers all ICT fields and is committed to providing a leading talent cultivation architecture and certification standards, cultivating ICT professionals in the digital era, and building a healthy ICT talent ecosystem.

Huawei Certification





Huawei Datacom Certification Portfolio



HCIA-Datacom

Aiming to cultivate network engineers with basic datacom theories and skills

HCIE-Datacom

Aiming to cultivate network experts with solid theoretical foundation and deployment capabilities for cross-field solutions

HCIP-Datacom

Aiming to cultivate senior network engineers for cross-field solution planning and design or single-field planning and deployment



Quiz

1. (Single) Which of the following type of network topology has the highest reliability? ()
 - A. Star network topology
 - B. Ring network topology
 - C. Full-mesh network topology
 - D. Tree network topology



Summary

- This section describes the concepts of network communication and data communication network. The basic function of a data communication network is to implement network communication.
- This section also introduces various network devices, the differences between LAN, MAN and WAN, and various network topologies. In actual networking, multiple topologies are combined according to the requirements of multiple parties.
- This section also describes network engineering and network engineers and introduces Huawei datacom certification system.

The background of the slide features a blue-tinted image of a modern office interior. In the foreground, several groups of business professionals are silhouetted against the bright light coming from large windows. They appear to be in various stages of collaboration, some looking at documents or devices. The windows in the background offer a view of a dense urban skyline with numerous skyscrapers. The overall atmosphere is professional and high-tech.

Thank You
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